

written assignment

(1)

$$\nabla_{\theta} \text{Loss} = \left(\frac{-y}{h_{\theta}(x)} + \frac{1-y}{1-h_{\theta}(x)} \right) \nabla_{\theta} h_{\theta}(x) = \frac{h_{\theta}(x) - y}{h_{\theta}(x)(1-h_{\theta}(x))} \nabla_{\theta} h_{\theta}(x) = (\sigma(\theta^T x) - y)x.$$

So the SGD algorithm gives

$$\theta_{n+1} = \theta_n - \eta(\sigma(\theta^T x) - y)x = \theta_n + \eta(y - \sigma(\theta^T x))x.$$



想問，這個符號是什麼